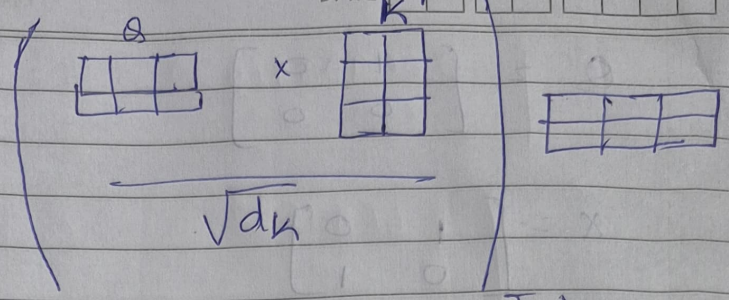
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$Z = \text{Softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) V$



$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) V$

$= \begin{bmatrix} & & \\ & & \end{bmatrix}_{2 \times 3}$ //

$x_1 \quad x_2$

x = bank river

$x_1 = \begin{bmatrix} 1.0 \\ 0.0 \end{bmatrix}$ $\quad x_2 = \begin{bmatrix} 0.0 \\ 1.0 \end{bmatrix}$

$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ $\quad W_K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $\quad W_V = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$

$\text{Attention}(Q, K, V) = \begin{bmatrix} & \\ & \end{bmatrix}_{2 \times 2}$ $\rightarrow z_1$ for bank

$\rightarrow z_2$ for river

Scaled dot prod attention

$x_1 \times W_Q = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

$Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$x_1 \times W_K = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

$x_1 \times W_V = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$

$= \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$

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$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow$$

$$X \times W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \downarrow$$

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$X \times W_K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$X \times W_V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$V = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \rightarrow$$

$$Z = \text{Softmax}\left(\frac{\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}{\sqrt{2}} \right) \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \text{Softmax}\left(\frac{\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}}{\sqrt{2}} \right) \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

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$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}} \qquad \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ \sqrt{2} & \sqrt{2} \end{pmatrix}$$

$$= \text{Softmax}\left(\frac{\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}}{\sqrt{2}} \right) \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \text{Softmax}\left(\frac{\begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}}{\sqrt{2}} \right) \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \text{Softmax}\left(\begin{bmatrix} 0.7071 & 0 \\ 1.4142 & 1.4142 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \right) \downarrow$$

$$= \text{Softmax} \begin{bmatrix} 0.7071 & 0 \\ 1.4142 & 1.4142 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} \text{Softmax}(0.7071 \quad 0) \\ \text{Softmax}(1.4142 \quad 1.4142) \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{e^{0.7071}}{e^{0.7071} + e^{0}} & \frac{e^{0}}{e^{0.7071} + e^{0}} \\ \frac{e^{1.4142}}{e^{1.4142} + e^{1.4142}} & \frac{e^{1.4142}}{e^{1.4142} + e^{1.4142}} \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2.0281}{3.0281012287} & \frac{1}{3.0281012287} \\ \frac{4.1137945937}{8} \approx 0.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

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$$= \begin{bmatrix} 0.6697596437 & 0.3302399505 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1.6697592 & 0.99071985 \\ 1.5 & 1.5 \end{bmatrix}$$